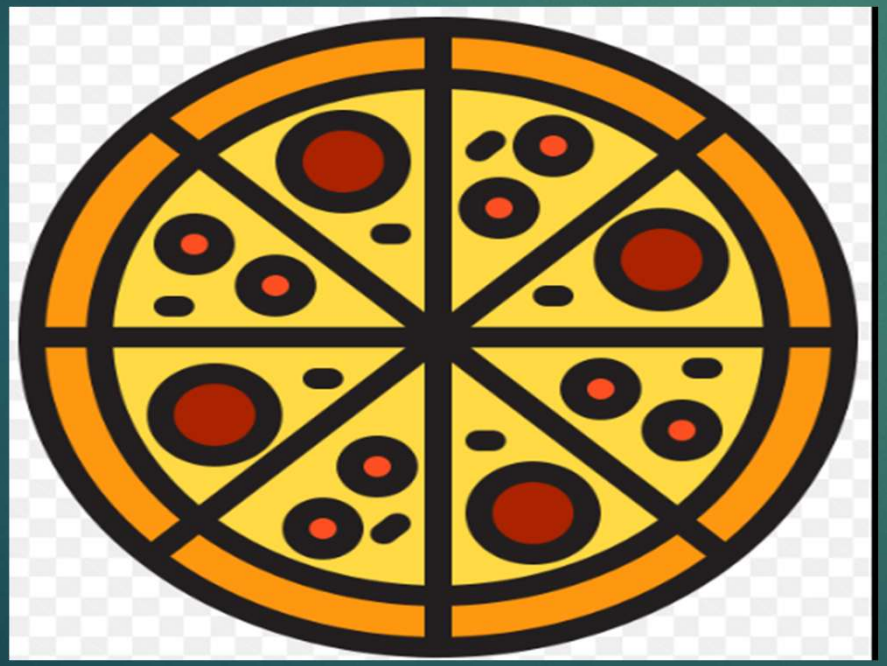
Pizza Sales Analysis Project

- Data Analysis Project using Excel, SQL, and Power BI



Project Overview

- Analyzed pizza sales data to uncover business insights and trends.
- This project focuses on analyzing pizza sales data to uncover meaningful business insights.
  The goal was to understand customer behavior, identify top-performing products, and optimize sales strategies using data-driven decisions.
- The analysis was performed using **Excel for data cleaning**, **SQL for querying**, and **Power BI for visualization**, resulting in an interactive dashboard that highlights key performance metrics.

Tools Used

- Excel
- SQL
- Power BI

Dataset

- Includes pizza sales records such as order date, quantity, category, and revenue.
- The dataset contains detailed records of pizza sales transactions, including:
- • Order ID and Date
  • Pizza Name and Category
  • Quantity Sold
  • Unit Price and Total Price
  • Size and Ingredients
- This structured dataset enables analysis of sales performance, customer preferences, and revenue trends.

Excel Analysis

- Performed data cleaning, pivot tables, and initial analysis.
- Performed data cleaning and preprocessing using Excel by handling missing values, correcting data types, and removing duplicates.
- Used Pivot Tables and Charts to analyze:
  - Sales by pizza category and size
  - Daily and monthly sales trends
  - Total revenue and quantity sold
- This step helped in understanding the dataset before advanced analysis.

SQL Analysis

- Used queries to extract insights like top-selling pizzas and revenue trends.

SQLQuery2.sql...n kumar (94))

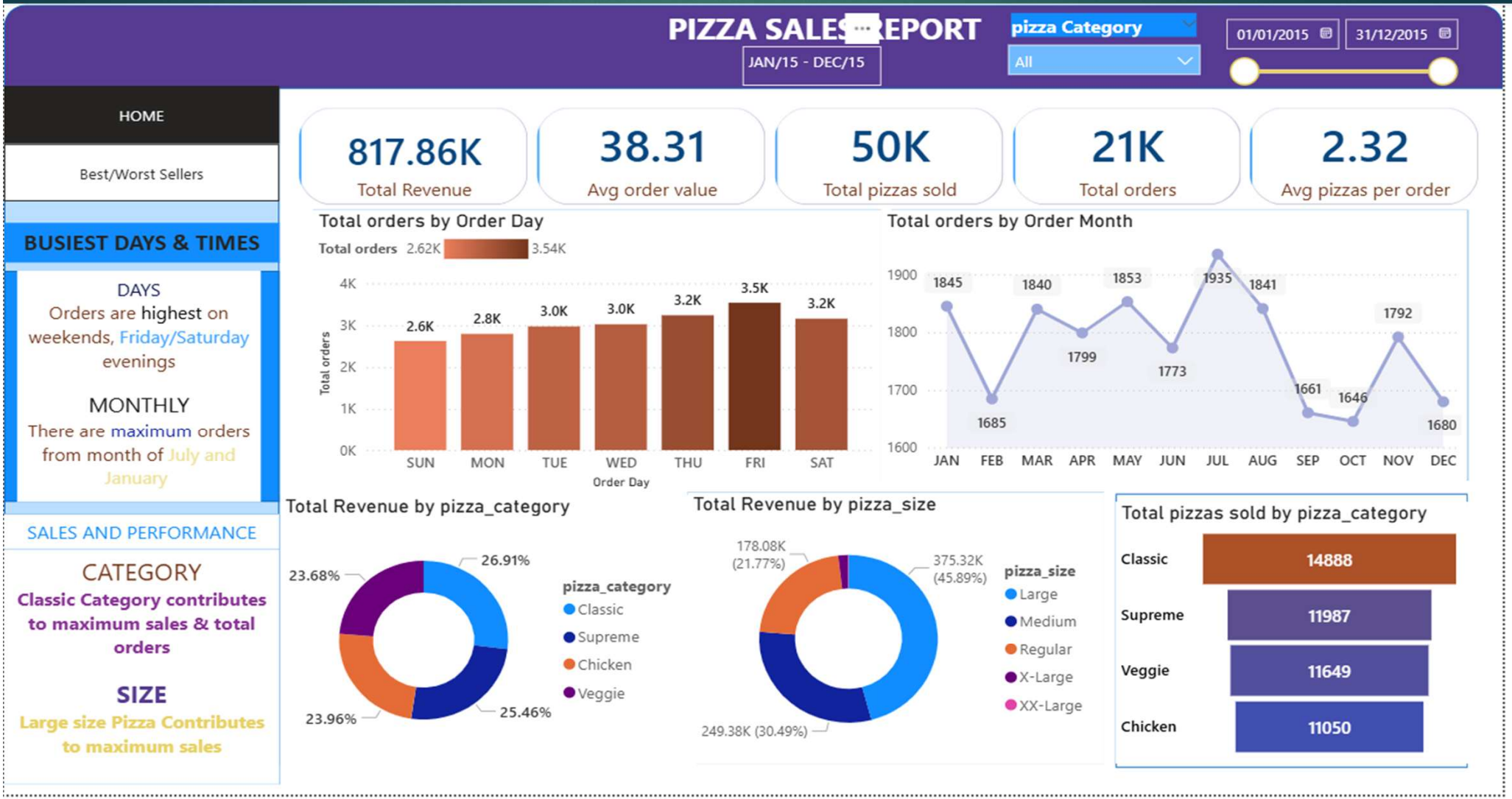
```
SELECT SUM(total_price) AS Total_Revenue FROM pizza_sales;

 Average Order Value
SELECT (SUM(total_price) / COUNT(DISTINCT order_id)) AS Avg_order_Value FROM pizza_sales

Total Pizzas Sold

SELECT SUM(quantity) AS Total_pizza_sold FROM pizza_sales

 Total Orders

SELECT COUNT(DISTINCT order_id) AS Total_Orders FROM pizza_sales



Average Pizzas Per Order

SELECT CAST(CAST(SUM(quantity) AS DECIMAL(10,2)) /
CAST(COUNT(DISTINCT order_id) AS DECIMAL(10,2)) AS DECIMAL(10,2))
AS Avg_Pizzas_per_order
FROM pizza_sales


Daily Trend for Total Orders

SELECT DATENAME(DW, order_date) AS order_day, COUNT(DISTINCT order_id) AS total_orders
FROM pizza_sales
GROUP BY DATENAME(DW, order_date)

 Monthly Trend for Orders

select DATENAME(MONTH, order_date) as Month_Name, COUNT(DISTINCT order_id) as Total_Orders
from pizza_sales
GROUP BY DATENAME(MONTH, order_date)
```

```
% of Sales by Pizza Category

SELECT pizza_category, CAST(SUM(total_price) AS DECIMAL(10,2)) as total_revenue,
CAST(SUM(total_price) * 100 / (SELECT SUM(total_price) from pizza_sales) AS DECIMAL(10,2)) AS PCT
FROM pizza_sales
GROUP BY pizza_category

% of Sales by Pizza Size

SELECT pizza_size, CAST(SUM(total_price) AS DECIMAL(10,2)) as total_revenue,
CAST(SUM(total_price) * 100 / (SELECT SUM(total_price) from pizza_sales) AS DECIMAL(10,2)) AS PCT
FROM pizza_sales
GROUP BY pizza_size
ORDER BY pizza_size

Total Pizzas Sold by Pizza Category

SELECT pizza_category, SUM(quantity) as Total_Quantity_Sold
FROM pizza_sales
WHERE MONTH(order_date) = 2
GROUP BY pizza_category
ORDER BY Total_Quantity_Sold DESC

Top 5 Pizzas by Revenue

SELECT Top 5 pizza_name, SUM(total_price) AS Total_Revenue
FROM pizza_sales
GROUP BY pizza_name
ORDER BY Total_Revenue DESC
```

```sql
Bottom 5 Pizzas by Revenue

SELECT Top 5 pizza_name, SUM(total_price) AS Total_Revenue
FROM pizza_sales
GROUP BY pizza_name
ORDER BY Total_Revenue ASC

Top 5 Pizzas by Quantity

SELECT Top 5 pizza_name, SUM(quantity) AS Total_Pizza_Sold
FROM pizza_sales
GROUP BY pizza_name
ORDER BY Total_Pizza_Sold DESC

SELECT TOP 5 pizza_name, SUM(quantity) AS Total_Pizza_Sold
FROM pizza_sales
GROUP BY pizza_name
ORDER BY Total_Pizza_Sold ASC

Top 5 Pizzas by Total Orders

SELECT Top 5 pizza_name, COUNT(DISTINCT order_id) AS Total_Orders
FROM pizza_sales
GROUP BY pizza_name
ORDER BY Total_Orders DESC


Borrom 5 Pizzas by Total Orders

SELECT Top 5 pizza_name, COUNT(DISTINCT order_id) AS Total_Orders
FROM pizza_sales
GROUP BY pizza_name
ORDER BY Total_Orders ASC
```

Power BI Dashboard

- Built interactive dashboard to visualize KPIs and trends.



PIZZA SALES REPORT

JAN/15 - DEC/15

pizza Category: All

Select all

The Barbecue Chicken ...

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Best/Worst Sellers

817.86K	38.31	50K	21K	2.32
Total Revenue	Avg order value	Total pizzas sold	Total orders	Avg pizzas per order

BEST SELLERS

REVENUE

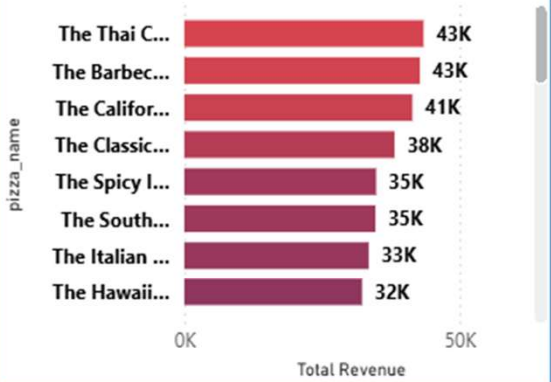
The thai chicken Pizza contributes to maximum Revenue

QUANTITY

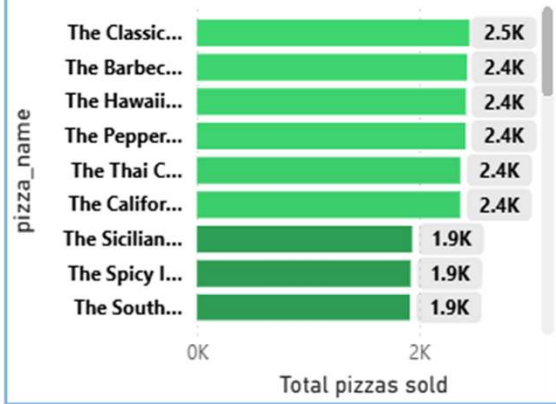
The classic Deluxe Pizza Contributes to maximum Total Quantities

TOTAL ORDERS

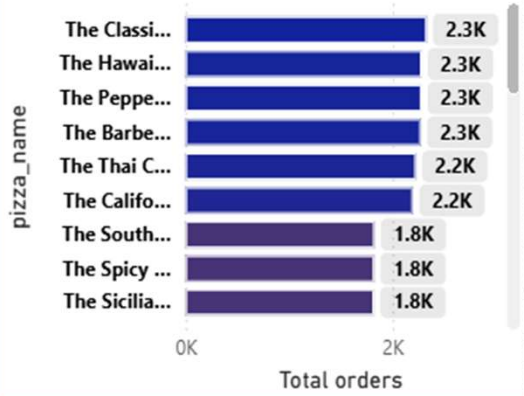
Top 5 pizzas by Revenue



Top 5 pizzas by Quantity



Top 5 pizzas by Total orders



WORST SELLERS

REVENUE

The Brie Carre minimum Revenue

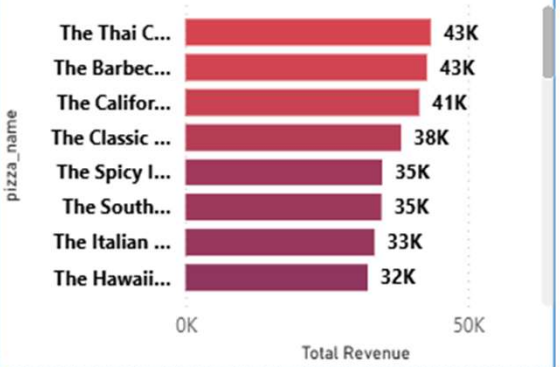
QUANTITY

The Brie Carre Pizza contribution to minimum Total Quantities

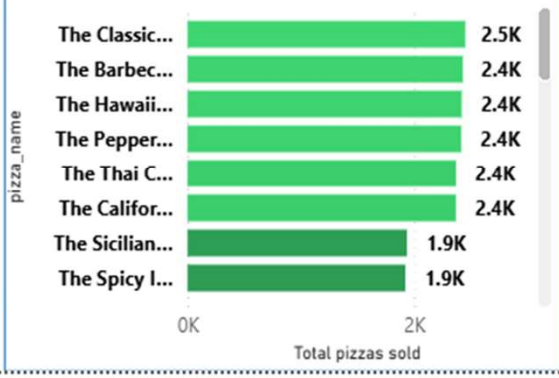
TOTAL ORDERS

The Brie Carre Pizza Contribution To minimum Total Orders

Top 5 pizzas by Revenue



Bottom 5 pizzas by Quantity



Top 5 pizzas by Total orders

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Key Insights

- • Best-selling pizza categories
- • Peak sales times
- • Revenue patterns

Conclusion

- Project demonstrates strong data analysis and visualization skills.
- This project highlights my ability to perform end-to-end data analysis, from data cleaning to visualization.
- Through this analysis, I identified key sales drivers, customer preferences, and revenue patterns that can help improve business performance.
- It reflects my skills in **data analysis, problem-solving, and dashboard creation**.

Thank You